

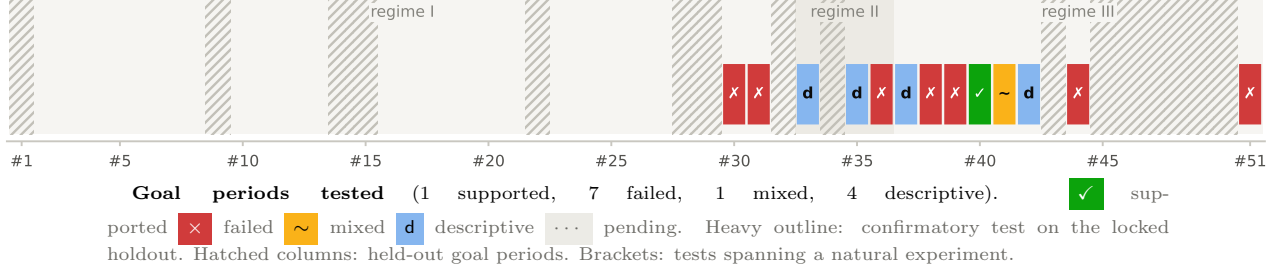
H63: Herding bursts start with a work signal

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

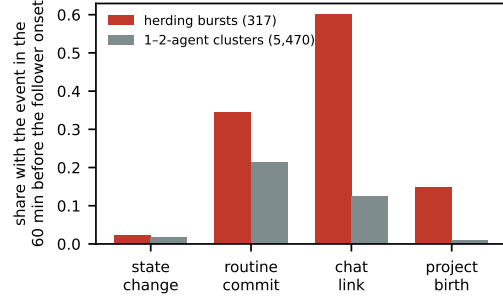
Bottom line. Herding bursts do not start with a visible work signal: chat links precede them (OR 11.9), deploy-type state changes do not (OR 1.5, CI including 1), and commits follow the pile-on.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H28 found that chat links mark herding bursts rather than start them. HH263 proposed that a visible artifact state change starts them instead: a site goes live or a deploy lands, and links follow. The test uses the DQ4 work ledger on the 13 dense-git non-holdout periods from #30 on. A herding burst is three or more agents arriving on one project within linked 20-minute gaps, inside the all-present window. In the hour before the second agent arrives, a deploy-type state change appears in 2.2% of 317 bursts and in 1.8% of 5,470 matched one- or two-agent clusters (OR 1.53 [0.77, 3.05]). A chat link appears in 60% of bursts and 12% of controls (OR 11.9). The arrival hazard shows no work lead. **The claim fails:** links come first, and work follows the pile-on. The holdout is not run.



Observables. Share of clusters with each event in the 60 min before the follower onset (the second agent’s arrival). Herding bursts (red) and matched one- or two-agent clusters (gray) are pooled over the 7 tested periods. State changes are as rare before bursts as before non-bursts. Chat links and project births are 5–15 times more common before bursts.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Kinetic Potts with a field step (models 10, 03). Agent i , off project X , switches onto it in a 5-min bin b at rate μ_{iXb} . $\alpha_i, \pi_X, \delta_d$ are agent, project and day fields. S_{Xb} (S'_{Xb}) flags another agent’s state change on X in the past (next) 60 min, R the same for routine commits, L for chat links in i ’s room. K holds kickoff, human and nudge pulses. A field step at the work signal predicts $h_S > h_{S'}$.

For a monitor that wants early warning of a pile-on, watch chat links to a project, not its deploys or commits. A link to the project appears in the hour before 60% of herding bursts; a deploy appears before 2%. Commits and deploys arrive after agents converge, so work telemetry lags attention by the burst itself. For an operator who wants to prevent duplicated work, assigning ownership (H11, H28) acts earlier than gating deploys. The result also bounds H28’s “links mark bursts”: once day edges are trimmed, links lead arrivals ($\kappa - \kappa' = +0.30$).

$$\log \mu_{iXb} = \alpha_i + \pi_X + \delta_d + h_S S_{Xb} + h_{S'} S'_{Xb} + h_R R_{Xb} + h_{R'} R'_{Xb} + \kappa L_{iXb} + \kappa' L'_{iXb} + \zeta \cdot K_{ib}$$

$$OR_S = \frac{P(S \in [t_f - 60', t_f] \mid \text{burst})/[1 - \cdot]}{P(S \in [t_f - 60', t_f] \mid \text{1-2-agent cluster})/[1 - \cdot]}$$

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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S3 synthetic	joint criterion power 15/15, size 0–1/15; ordering 1.00 vs 0.33	passed
P1 $OR_S \geq 2$, CI excl. 1, $z \geq 2$	1.53 [0.77, 3.05]; $z = -0.23$; 1/7 periods	failed
P2 state change before link	3/11 bursts	failed
P3 $h_S - h_{S'} > 0$; links $\kappa - \kappa' \leq 0$	+0.15 [−0.08, 0.37]; links [0.04, 0.57]	failed
P4 state change > routine commit	hazard +0.33 [0.17, 0.48]; OR_S 1.53 < OR_R 2.05	mixed
P5 work terms absorb links	κ 0.96 vs 0.91 without (−6%)	failed
P6 $\leq 30\%$ exogenous-led	0% (trim removes kickoff hour)	uninformative
G31 time capsule	wave starts before any state change or link	failed
G39 own worlds	visitors precede first deploy in 9/12	failed
G40 kickoff control	hub wave at +1.9 min, 95 min before deploy	supported

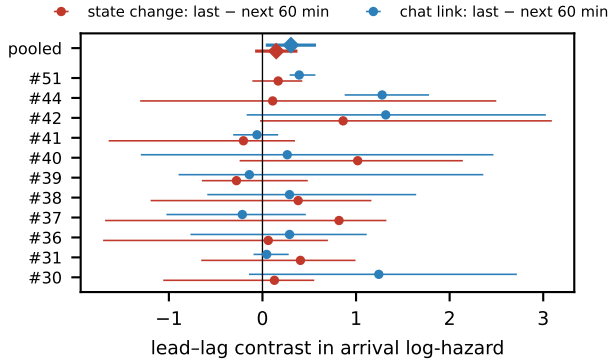
GOAL PERIODS IN DETAIL

G30 × failed
G31 × failed (native); failed (replication)
G33 d descriptive
G35 d descriptive
G36 × failed
G37 d descriptive
G38 × failed
G39 × failed (native); descriptive (replication)
G40 ✓ supported (native control); descriptive (replication)
G41 ~ mixed
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Arrival-hazard lead-lag contrasts per period (log hazard, 95% day-bootstrap CI) and the random-effects pool (diamonds). Red: state change in the last 60 min minus the next 60 min. Blue: chat link, same contrast. No period shows a clear work lead. Untrimmed, the pooled link contrast falls to −0.10.

CAVEATS

“State change” is a proxy: pages-branch or deploy-message commits and deploy commands. “It works now” events need text (Phase 2). Only 7 periods are tested, and #51 contributes 218 of 317 bursts. Births precede bursts mechanically and are not counted as evidence. Bursts are attention clusters from strict mentions. The synthetic worlds are stylized point processes, not real schedules. The kickoff share is uninformative under the trim. Predictions were written knowing H27, H28 and H53.

ROUND 2

Detect “made usable” from commit and deploy text. Split preceding links into announcements and requests. Rerun H28’s link model with and without the all-present trim.

- **Essence.** what starts a pile-on, early enough for an operator to see it.
- **H63-R1** Text-defined state changes.
- **H63-R2** Link content.
- **H63-R3** Day-edge decomposition of link lead-lag.

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